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Configuring SSH Remote Access on Ubuntu Server

Objective

Set up an Ubuntu Server with SSH (Secure Shell) remote access and connect to it from a host desktop machine using Bridged Network mode.

Operating System Installation

Install Ubuntu Server

1. Start the VM and boot from the Ubuntu Server ISO
2. Follow the installation wizard:
 - Profile setup: Create a user account (e.g., server / server123)
 - Install OpenSSH server: **Check this option** (Important!)
 - Selected software: **None** (Standard Ubuntu Server)
3. Wait for installation to complete and reboot

Network Configuration

Configure Network Mode

For VirtualBox:

1. Power off the VM (if running)
2. Go to **Settings** → **Network**
3. **Adapter 1**: Enable Network Adapter
4. **Attached to**: Select **Bridged Adapter**
5. **Name**: Select your host's physical network adapter (Wi-Fi/Ethernet)
6. Click **OK**

For VMware:

1. Power off the VM
2. Go to **VM Settings** → **Network Adapter**

3. Select **Bridged** mode

4. Click **OK**

Install/Verify SSH Server

Server: Install SSH on the server (if not already installed):

- `sudo systemctl status ssh` – for checking the SSH status.
- `sudo apt update` – updating the your server.
- `sudo apt install openssh-server -y` –installing the SHH.
- `sudo systemctl enable ssh` – enabling the SSH to your unit. (SERVER/DESKTOP)
- `sudo systemctl start ssh` – starting the SHH to your unit. (SERVER/DESKTOP)

Bridged makes the server behave like a separate PC in your LAN.

1. Find the Server's IP

Inside the VM, run:

```
Sample@UBUNTU-DESKTOP:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:9b:8c:ca brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.4/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s3
        valid_lft 83602sec preferred_lft 83602sec
    inet6 2001:4451:b00:2000:d76c:1af2:a10c:f36d/64 scope global temporary dynamic
        valid_lft 179068sec preferred_lft 83386sec
    inet6 2001:4451:b00:2000:a00:27ff:fe9b:8cca/64 scope global dynamic mngtmpadr proto kernel_r
        valid_lft 179068sec preferred_lft 92668sec
    inet6 fe80::a00:27ff:fe9b:8cca/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
Sample@UBUNTU-DESKTOP:~$
```

From the desktop, connect to the server: `ssh username@192.168.1.120`

2. Connect from Desktop

From your host desktop:

```
Sample@UBUNTU-DESKTOP:~$ ssh Server@192.168.1.14
Server@192.168.1.14's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-79-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Sep  8 11:24:38 PM UTC 2025

System load:            0.0
Usage of /:             5.1% of 48.91GB
Memory usage:          5%
Swap usage:            0%
Processes:             139
Users logged in:       1
IPv4 address for enp0s3: 192.168.1.14
IPv6 address for enp0s3: 2001:4451:b00:2000:a00:27ff:fe57:bc4a

Expanded Security Maintenance for Applications is not enabled.

136 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Sep  8 22:52:07 2025 from 192.168.1.4
Server@UBUNTU-SERVER:~$
```

Submission Requirements

1. Screenshots (labeled clearly):

- VM network settings showing Bridged Adapter
- Server IP address (output of `ip a` or `hostname -I`)
- SSH service status (`sudo systemctl status ssh`)
- Successful SSH connection from host
- Verification commands executed inside VM

2. Submission Format:

- Single PDF file
- Include name, student ID, and date
- All screenshots with brief explanation
- Upload to your portfolio website.

UBUNTU SERVER SSH REMOTE ACCESS LAB

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Course/Year/Section: BSIT3B

Date Submitted: 02-09-26

Instructor:Johneros P. Puyo

1 OBJECTIVES (5 POINTS)

Check all that apply:

- ☐ **Create and configure a Virtual Machine with Ubuntu Server**
- ☐ **Configure Bridged Network Adapter for network access**
- ☐ **Determine and verify the server's IP address**
- ☐ **Install OpenSSH Server and enable the service**
- ☐ **Establish successful SSH connection from host machine**
- ☐ **Execute verification commands to confirm remote access**

2 HARDWARE & SOFTWARE REQUIREMENTS (5 POINTS)

Item	Specification
Host Operating System	Oracle Linux
Virtualization Software (VirtualBox/VMware)	Vm Virtual Bos
Ubuntu Server Version	Ubuntu 26.04 LTS
VM Name	BubanElectiveServer
RAM Allocated	4
Storage Allocated	10048
Network Mode	Host Only Adapter
SSH Client Used	Secure shell client

3 VM CONFIGURATION

Setting	Value
VM Name	UBUNTU-SERVER
OS Type	Ubuntu (64-bit)
Memory	10048 MB
Hard Disk	108 GB
Network Adapter	Bridged

4. NETWORK CONFIGURATION

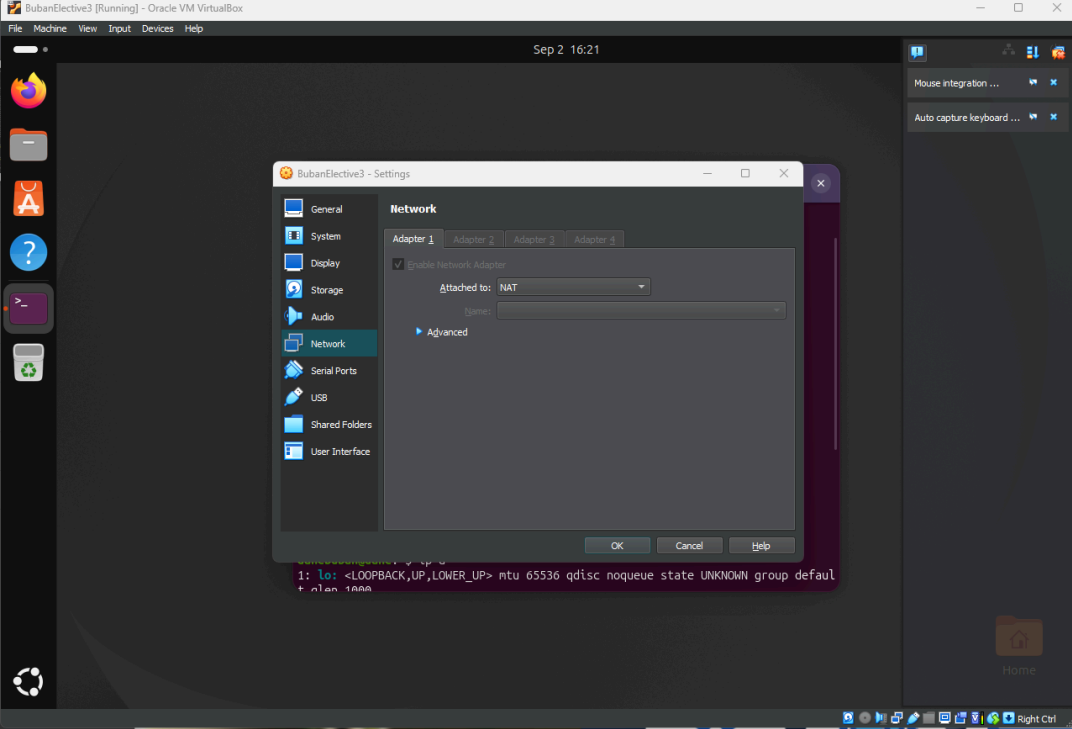
Commands Executed:

Server IP Address: 192.168.58.101

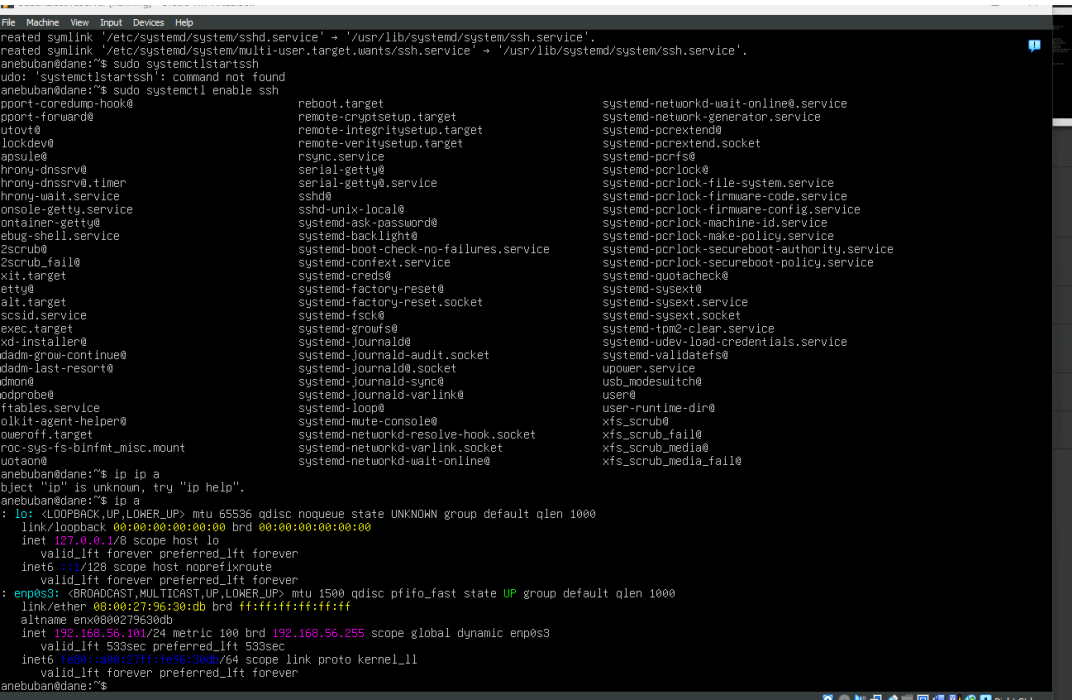
Network Interface: Host-only Adapter

6. REMOTE CONNECTION Connection Details:

Parameter	Value
Server IP	192.168.58.101



I putty my network status into nat



After i login and put command ip a so that i can see my ip address

```
inet6 fe80::a00:27ff:fe98:300b/64 scope link proto kernel llmnr
valid_lft forever preferred_lft forever
anebuban@dane:~$ sudo systemctl status ssh
sudo: authenticate] Password:
ssh.service - OpenBSD Secure Shell server
Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
Active: active (running) since Wed 2026-09-02 06:34:21 UTC; 1h 52min ago
Invocation: 3946fab5cabe4dcea8677566973e1f9c
TriggeredBy: ● ssh.socket
Docs: man:sshd(8)
      man:sshd_config(5)
Main PID: 1374 (sshd)
Tasks: 1 (limit: 8196)
Memory: 5.2M (peak: 7.5M)
CPU: 211ms
CGroup: /system.slice/ssh.service
        └─1374 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
```

Then i put a command ssh so if its working



```
bunantictiveServer [running] - Oracle VM VirtualBox
File Machine View Input Devices Help
buntu 26.04 LTS dane tty1
ane login: danebuban
assword:
elcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Wed Sep  2 08:28:04 AM UTC 2026

System load: 0.0           Memory usage: 3%   Processes:    151
Usage of /:  13.8% of 52.07GB Swap usage:   0%   Users logged in: 0

xpanded Security Maintenance for Applications is not enabled.

3 updates can be applied immediately.
o see these additional updates run: apt list --upgradable

nable ESM Apps to receive additional future security updates.
ee https://ubuntu.com/esm or run: sudo pro status

anebuban@dane:~$
```

Welcome

```
danebuban@dane: ~ — ssh danebuban@192.168.56.1...
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Wed Sep  2 07:05:22 2026 from 192.168.56.1
danebuban@dane:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP
   group default qlen 1000
   link/ether 08:00:27:96:30:db brd ff:ff:ff:ff:ff:ff
   altname enx0800279630db
   inet 192.168.56.101/24 metric 100 brd 192.168.56.255 scope global dynamic en
   p0s3
       valid_lft 459sec preferred_lft 459sec
   inet6 fe80::a00:27ff:fe96:30db/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
danebuban@dane:~$ =
```

The i put a command to my terminal in desktop ubuntu to connect it.